

Eli Foster

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Education

Bucknell University, Lewisburg PA

2023-2027

- Bachelor's in Electrical Engineering, 3.65 GPA/3.77 Engineering GPA, graduation **May 2027**
- Concentration in semiconductor technology through several classes and research projects
- Member and former VP of IEEE, executive board member of Bucknell Baja SAE

Employment

EOS Energy – *Electrical Engineering Intern, Battery Management Systems* June 2026-August 2026

- Electronics R&D team intern responsible for creating effective engineering documentation, verification of new designs, extensive testing and debugging, and design of prototypes
- Helped bring up new board designs from ideation into manufacturing in large-scale production lines, ensuring design is kept accurate and effective. Responsible for some schematic and layout of PCBs working with BMS's up to several kilovolts in capacity, requiring precise design
- Conducted thorough validation and verification of prototype designs, using benchtop equipment including oscilloscopes and power supplies, as well as creating thorough documentation for CE certification. Followed agile methodology using Jira

Bucknell University – Multiple Roles

May 2024–Present

Maker-E Project Developer

(Summer 2024 & Summer 2025)

- Expanded and optimized the PortalBox embedded hardware platform for improved makerspace equipment access, focusing on analog and digital circuit layout and PCB design
- Conducted firmware testing, bug identification, and feature enhancements in collaboration with faculty and technical IT staff
- Created educational technology kits to introduce students to modern engineering tools, focusing on microcontrollers, PCB design, computer architecture, and manufacturing equipment

Makerspace Manager

(Academic Years 2024–2027)

- Maintained and repaired 3D printers, laser cutters, PCB assembly machines, and lab equipment
- Certified users on manufacturing equipment and provided training on best practices for electronics fabrication and testing
- Assisted users with EE core concepts including circuit design and layout, Python, and similar. TA for several ECE classes covering Assembly, C, and core concepts

Projects

RFID Makerspace Equipment Access

Summer 2025

- Completed v5.0 of an open-source makerspace project known as the Portalbox
- Focused on embedded hardware to create an interlock system for both power and data, using KiCAD and Altium Designer
- An ESP32-S3 Devkit based system checked in with a locally hosted SQL database to verify user access, using firmware written in MicroPython designed after v4.0 firmware written in Python
- Utilized Git for version control, Embedded Linux to build and flash firmware, and software validation to ensure integrity of modules and libraries
- Thoroughly tested and debugged PCB once populated using oscilloscopes, multimeters, and microscopes. Ensured full functionality before continuing the project

Bucknell Baja SAE

Academic Year 24-25

- Led a team of ~10 ECE engineers in order to create all electrical systems on a racing vehicle
- Acted as client for ECE Senior Design team, supplying requirements, specifications and feedback
- Implemented a wiring harness, electronic differential controls, and an instrumentation system to monitor vehicle status and statistics
- Utilized Agile-style development, created technical documentation, and R&D skills